Instability #21a: Closed

$T_{\text{avg}} = 2nK^*/[(n-1)\epsilon_0]$ Derived from δ-Chain
Gap 1.2% from V20.3. No Free Parameters.

The drift rate of D_{KL} in Phase I is $\mu = \epsilon_0(n-1)/(2n)$. Wald's formula gives $T_{\text{avg}} = K^*/\mu$. All quantities from δ.

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Abstract

Instability #21a asked: derive T_{avg} (mean inter-event time) from the δ-chain. This document closes it.

The derivation: In Phase I (between R-events), the mean D_{KL} between a node and its neighbours grows at rate:

$$\mu = \frac{\epsilon_0(n-1)}{2n}$$

This is the drift of a Dirichlet process under background noise ϵ_0, with n dimensions.

By Wald's identity for drift-dominated first-passage:

$$T_{\text{avg}} = \frac{K^*}{\mu} = \frac{2nK^*}{(n-1)\epsilon_0}$$

Numerically: $\mu = 0.00755 \times 7/16 = 0.003303$ per sweep. $T_{\text{avg}} = 0.1170/0.003303 = 35.42$ sweeps. V20.3 measured: $T_{\text{avg}} \approx 35$ sweeps. Gap: **1.2%**.

Derivation chain (no free parameters): $\delta \to \epsilon_0$ (EIC SINDy) $\to \delta \to n = 8$ (SO(3), #16) $\to \delta \to K^* = 0.1170$ (θ-formula) $\to \mu = \epsilon_0(n-1)/(2n) \to T_{\text{avg}} = K^*/\mu = 35.4$.

Consequence for β: With T_{avg} derived, H_{before} is determined from the entropy ODE, giving $\Delta H = 2.040$. The scale factor $s = \beta\kappa/\Delta H = 0.163$ is the only remaining open quantity (#21b).

Status: #21a **CLOSED**. #21b open. #21 requires #21b.

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1 The Drift Rate of D_{KL} in Phase I

Drift Rate μ from Dirichlet Noise

In Phase I, a node's distribution P_i evolves under background noise:

$$P_i(t+1) = (1-\epsilon_0)\, P_i(t) + \epsilon_0 \,\mathrm{Dir}(\mathbf{1}_n)$$

Similarly for neighbours P_j.
The mean $D_{\mathrm{KL}}(P_i \| P_j)$ between two independently ϵ_0-noised Dirichlet vectors grows at rate:

$$\mu = \frac{d}{dt}\,\mathrm{E}[D_{\mathrm{KL}}(P_i \| P_j)] = \epsilon_0 \cdot \frac{n-1}{2n}$$

Derivation: for symmetric $\mathrm{Dir}(\mathbf{1}_n)$, $\mathrm{E}[D_{\mathrm{KL}}(\mathrm{Dir}\|\mathrm{Dir})] = (n-1)/n$. Each sweep adds ϵ_0 weight of fresh Dirichlet to each node. The net growth rate of cross-KL is half this (both nodes move):

$$\mu = \epsilon_0 \cdot \frac{n-1}{n} \cdot \frac{1}{2} = \frac{\epsilon_0(n-1)}{2n}$$

2 Wald's Formula for First-Passage Time

$T_{\mathrm{avg}} = K^*/\mu$ by Wald's Identity

Phase I dynamics is drift-dominated: the drift-to-noise ratio is

$$\frac{2\mu K^*}{\sigma^2} = \frac{2\mu K^*}{\epsilon_0^2(n-1)/n} = \frac{(n-1)K^*/(n\epsilon_0)}{1/2} = \frac{(n-1)K^*}{n\epsilon_0/(n-1)/n}$$

Numerically: $2\mu K^*/\sigma^2 = 15.5 \gg 1$ (drift-dominated).
For a drift-dominated first-passage process starting at $D_{\mathrm{KL}} = 0$ and hitting barrier K^* with drift μ:

$$\boxed{T_{\mathrm{avg}} = \frac{K^*}{\mu} = \frac{2nK^*}{(n-1)\epsilon_0}}$$

All parameters derived:
- $n = 8$: from SO(3) (#16, closed)
- $\epsilon_0 = 0.00755$: from EIC SINDy ($|I_{\mathrm{eff}} - \ln 2|$)
- $K^* = 0.1170$: from θ-formula self-consistency (CRF_Tavg_Derivation)

No free parameters.

3 Numerical Verification

Verification Against V20.3

Quantity	Derived	V20.3
$\mu = \epsilon_0(n-1)/(2n)$	0.003303 per sweep	—
$T_{\text{avg}} = K^*/\mu$	**35.42 sweeps**	**35 sweeps**
Gap	**1.2%**	—

The 1.2% gap is consistent with approximations in the drift model (neighbour coupling, mass effects) that contribute corrections of order ϵ_0.

4 Consequence for β

β Chain with T_{avg} Now Derived

From the entropy ODE (CRF_HDynamics):

$$H_{\text{before}} = \frac{H_{\text{after}}}{1 + \alpha H_{\text{after}} T_{\text{avg}}}$$

where $H_{\text{after}} = \ln n = \ln 8 = 2.079$ (from Born rule) and $\alpha = \ln 2$ (from δ-bit). With $T_{\text{avg}} = 35.42$:

$$H_{\text{before}} = \frac{2.079}{1 + 0.693 \times 2.079 \times 35.42} = \frac{2.079}{52.05} = 0.040$$

$$\Delta H = H_{\text{after}} - H_{\text{before}} = 2.079 - 0.040 = 2.040$$

The β formula:

$$\beta = \frac{\Delta H \cdot s}{\kappa}$$

With $\beta_{\text{measured}} = 2.212$ and $\kappa = 0.15$:

$$s = \frac{\beta\kappa}{\Delta H} = \frac{2.212 \times 0.15}{2.040} = 0.163$$

$s \approx 0.163$ is the scale factor relating IDV units to ΔH units — this is #21b, still open.

5 The Full C Expression

C as a Derived Quantity

The shape factor C in $T_{\text{avg}} = C\exp(K^*/n\epsilon_0)$ is now:

$$C = \frac{T_{\text{avg}}}{\exp(K^*/n\epsilon_0)} = \frac{2nK^*}{(n-1)\epsilon_0} \cdot \exp\left(-\frac{K^*}{n\epsilon_0}\right)$$

Numerically: $C = 35.42/6.94 = 5.10$ (vs empirical 5.04, gap 1.3%).
C is *not* an independent shape parameter. It is a derived consequence of T_{avg} and the structural factorisation. The "shape factor" mystery is dissolved: C encodes the ratio of barrier height to drift length, modulated by the natural diffusion scale.

6 Updated Status

Result	**Status**	**Note**
$K^* = 0.1170$	Derived	θ-formula
$\mu = \epsilon_0(n-1)/(2n)$	**Derived**	Dirichlet drift
$T_{\text{avg}} = K^*/\mu = 35.4$	**Derived**	Wald's formula, 1.2%
$\Delta H = 2.040$	**Derived**	Entropy ODE
$C = 5.10$ (closed form)	**Derived**	From T_{avg}
$s = 0.163$ (#21b)	Open	Wave field scale factor
$\beta = d_s/(2\ln 2)$ formally	Open	Requires #21b
#21a	**CLOSED**	1.2% gap
#21b	Open	Next target
#21	Open	Requires #21b

$T_{\text{avg}} = 2nK^*/[(n-1)\epsilon_0]$.

n came from SO(3). SO(3) came from δ.
ϵ_0 came from EIC. EIC came from δ.
K^* came from θ. θ came from δ.

Wald's formula connected them.
The mean time between R-events is derived.

#21a is closed.